Movie Memory

Team RSS

March 10, 2026

1 Experiment Details

Participants: A total of 171 individuals participated, 13 were excluded from analysis due to a data error, leaving a final sample of $N = 146$ participants. All participants reported normal or corrected-to-normal vision, and the majority were right-handed. Participants were randomly assigned to one of two between-subjects conditions: the Natural Boundary (NB) condition or the Abrupt Boundary (AB) condition.
Stimuli: Forty YouTube Shorts were selected. To identify event boundaries in each clip, an independent group of annotators watched the videos and marked event boundaries.
Condition Manipulation: In the **NB** condition, videos played continuously to their natural conclusion. Event boundaries occurred exactly when viewers would expect them, preserving cognitive continuity. In the **AB** condition, videos were terminated *before* a consensus boundary and immediately resumed at the onset of the subsequent event. Video duration was matched across conditions to ensure it did not serve as a confounding factor.
Recognition Test Stimuli: For the recognition memory test, two types of target frames were extracted. **Before-Boundary (BB)** frames were still images captured just prior to an event boundary. **Event-Middle (EM)** frames were captured from the centre of an event, temporally distant from any boundary. For each target frame, a corresponding lure frame was selected.
Phase 1: Encoding: Participants were instructed to watch a series of short video clips. Five videos were repeated as vigilance checks. Participants who exceeded 27.05 minutes during the encoding phase were flagged as inattentive.
Phase 2: Recognition Memory Test: Following encoding, on each trial, two frames from the same video were presented side-by-side: one target and one lure frame. Response accuracy (`resp.corr`: 0 = incorrect, 1 = correct) and response time (`resp.rt`) were recorded. Immediately, participants rated their confidence on Likert scale (1 = *very unconfident*, 5 = *very confident*), recorded as `conf_radio.response`. Each participant completed approximately 40 recognition trials. In addition, two composite indices were computed, **Recognition Memory Index (REC)** and **Lure Discrimination Index (LDI)**.

2 Literature

Event Segmentation Theory (EST) posits that the perceptual system continuously parses the ongoing stream of experience into discrete, meaningful units referred to as events. Central to the theory is the notion of an *event model*: a working memory representation that guides perceptual processing by generating predictions about what will happen next. However, when prediction error increases when the environment becomes less predictable the current event model can no longer adequately account for new input [1]. Building directly on EST, [2] proposed the Event Horizon Model, and the *boundary advantage*. This principle holds that event boundaries serve as privileged anchor points in LTM. Recall and recognition of information presented at or near event boundaries tends to be superior to information from event interiors. A closely related finding reported by [3] demonstrated that deleting temporal segments from within events impairs memory less than deleting equivalent segments at event boundaries, directly demonstrating that boundaries represent the most cognitively critical moments in a continuous experience. Empirical support for the link between event boundaries and memory encoding in film contexts was provided early by [4]. In this study, participants viewed a feature film into which advertisement breaks were inserted at either natural event boundaries or at non-boundary locations within events. Advertisements inserted at natural boundaries were associated with improved subsequent recall of the surrounding film content, whereas advertisements inserted at non-boundary didn't.

The specific prediction that items presented near event boundaries are better encoded than items from event interiors was tested directly by [5]. Across all experiments, objects that were present at the time of boundary perception were subsequently recognised with higher accuracy than those presented within event interiors. Notably, this boundary advantage was observed even across the boundary that is, objects from the preceding event were still better remembered when a boundary followed shortly after their presentation. This finding maps directly onto the design of the present study. The BB frames are drawn from the temporal region immediately preceding an event boundary, the zone in which [5] observed the strongest encoding advantage. The EM frames, by contrast, are drawn from the event interior, where encoding advantage is minimal. The NB vs. AB manipulation then allows us to test whether the boundary advantage for BB frames depends on the natural boundary transition completing undisturbed, or whether it persists even when the boundary moment is artificially removed.

3 Descriptive Statistics

3.1 Sample Characteristics

The final analytical sample consisted of 146 participants who completed a total of 6,800 recognition memory trials. Demographic characteristics are summarised in Table 1.

Table 1: Sample Demographic Characteristics by Condition

Characteristic	**NB** ($n = 79$)	**AB** ($n = 67$)	**Total** ($N = 146$)
Age: M (SD)	22.2 (2.1)	21.9 (1.6)	22.0 (1.9)
Male / Female	63 / 16	49 / 18	112 / 34
Right-handed	75 (94.9%)	64 (95.5%)	139 (95.2%)
Normal vision	54 (68.4%)	46 (68.7%)	100 (68.5%)
Corrected-to-normal vision	25 (31.6%)	21 (31.3%)	46 (31.5%)

Recognition accuracy was high overall ($M = 0.857$, $SD = 0.077$), with NB participants outperforming AB participants (0.872 vs. 0.838). The AB condition showed greater variability and a lower minimum score, suggesting less consistent encoding under boundary disruption. Mean response time was 5.718 s ($Mdn = 5.470$ s); NB participants were slightly slower but more accurate than AB participants, inconsistent with a simple speed-accuracy tradeoff. Confidence followed the same pattern, with NB participants reporting higher mean confidence (4.21 vs. 4.08), and the alignment between accuracy and confidence indicates adequate metacognitive awareness. At the trial level, 5,822 of 6,800 trials (85.6%) were correct, with NB trials yielding 87.1% accuracy versus 84.0% for AB. Full statistics are reported in Tables 2.

Table 2: Descriptive Statistics by Condition

Measure	**Condition**	M	SD	Mdn	**Min**	**Max**
Accuracy	NB	0.872	0.069	0.875	0.675	0.988
	AB	0.838	0.084	0.850	0.575	0.975
	Overall	0.857	0.077	0.875	0.575	0.988
RT (participant, s)	NB	5.834	1.636	5.534	3.256	12.180
	AB	5.582	1.491	5.399	3.251	11.633
	Overall	5.718	1.571	5.470	3.251	12.180
Confidence (1–5)	NB	4.209	0.407	4.275	3.000	5.000
	AB	4.077	0.431	4.100	2.900	4.900
	Overall	4.149	0.422	4.225	2.900	5.000

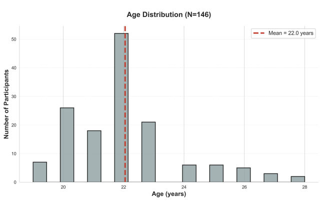
In summary, three consistent patterns emerge from the descriptive statistics. First, NB participants demonstrated superior recognition memory performance across all measures, higher accuracy (87.2% vs. 83.8%), higher confidence (4.21 vs. 4.08), and lower performance variability providing preliminary support for the boundary-specific encoding hypothesis. Second, the AB condition was associated with faster yet less accurate responses, suggesting differences in retrieval strategy rather than a simple speed-accuracy tradeoff.

4 Data Preprocessing

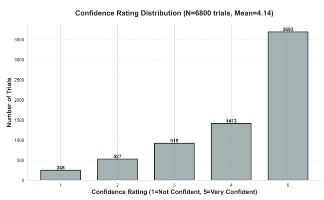
For data cleaning and preprocessing, response variables including accuracy (`resp.corr`), response time (`resp.rt`), and confidence ratings (`conf_radio.response`) were converted to numeric format.
We adopted *no-data-loss* approach for participant retention. All 146 participants who completed the recognition memory task were retained in the final sample, regardless of demographic data completeness. Among these participants, 38 individuals (26.0%) had missing demographic information in one or more variables. For the continuous variable age, missing values were imputed using the median value within each experimental condition (NB vs. AB). For categorical variables (gender, handedness, vision), missing values were replaced with the mode within each respective condition.

5 Data Visualisations

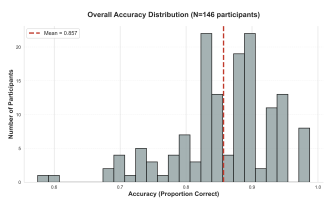
Method: Four distributional visualisations were produced. Age distribution was examined using a histogram with 20 equal-width bins. Confidence ratings were visualised as a bar chart, as discrete bars better respect the categorical nature of the data. Recognition accuracy was examined at the participant level using a histogram with 25 equal-width bins. Response time was characterised at the trial level ($N = 6800$) using a 60-bin histogram with both mean and median overlaid.



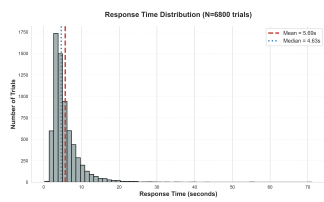
(a) Age



(b) Confidence Rating (Bar)



(c) Accuracy



(d) Response Time

Figure 1: Visualization across different analyses

Interpretation: The age histogram shows a right-skewed, multimodal distribution concentrated between 20 and 24 years, consistent with a university sample. The mean age line sits near the centre of the main cluster, confirming sample homogeneity. In confidence bar chart, the majority of responses clustered at ratings 4 and 5. This validates engagement with the task, the ceiling concentration reduces the sensitivity of confidence as a discriminating variable. The accuracy histogram shows a left-skewed distribution concentrated between 0.80 and 1.00. The response time histogram shows the characteristic strong positive skew of RT data, with the bulk of the 6,800 trials falling between 2 and 10 seconds but a long right tail extending beyond 20 seconds.

Method: Violin plots were used to compare distributions across conditions (AB vs. NB) and frame types (BB vs. EM) on response time, confidence, and accuracy. A final panel plots confidence stratified by response correctness.

Interpretation: The response time violin for AB vs. NB shows broadly overlapping distributions, but the NB violin is marginally wider at higher RT values. Both distributions are positively skewed with long upper tails. The confidence violin by condition shows near-identical distributions between AB and NB, both heavily concentrated at ratings 4 and 5. The NB violin shows a marginally higher density

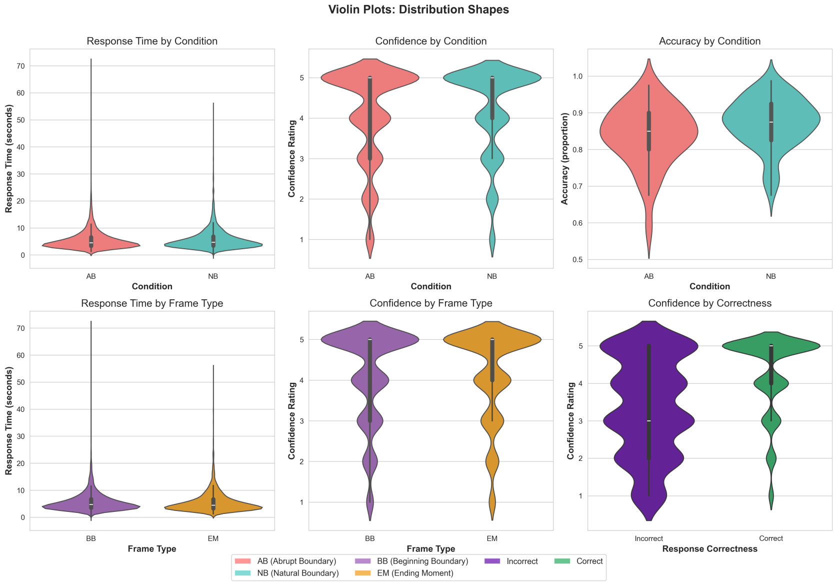


Figure 2: Violin plots

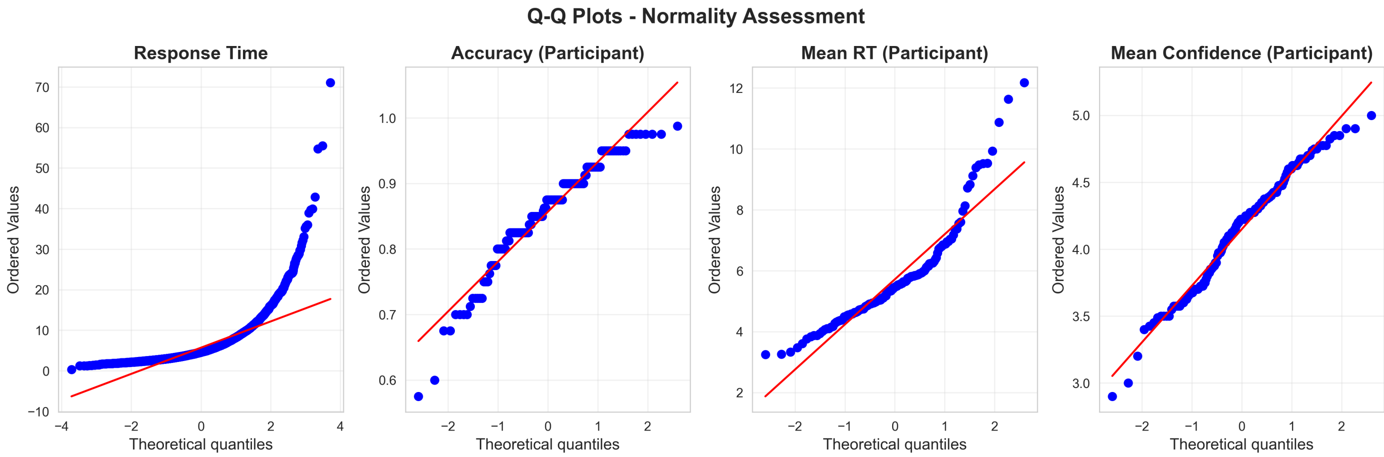


Figure 3: QQ plots

at rating 5. The accuracy violin by condition shows the NB distribution positioned slightly higher than AB, with the NB median box sitting above that of AB. More importantly, the AB violin has a more pronounced lower tail. It suggests a subset of AB participants who were meaningfully impaired by the boundary disruption. The response time violin by frame type (BB vs. EM) shows largely overlapping distributions. The confidence violin by frame type again shows strong overlap, with both BB and EM distributions concentrated at ratings 4–5. The confidence by correctness violin is the most theoretically informative panel. This confirms that participants were more confident when they were right than when they were wrong.

Method and Interpretation: Normality was assessed for four variables using Q-Q plots alongside three formal tests: the Shapiro-Wilk test, the Kolmogorov-Smirnov (KS) test, and the D'Agostino-Pearson test. All four variables departed significantly from normality. Trial-level response time was non-normal across all three tests, consistent with the strong positive skew typical of RT distributions. Participant-level mean RT was similarly non-normal. Participant-level recognition accuracy also violated normality, reflecting a negatively skewed distribution with performance concentrated near ceiling. Mean confidence per participant showed borderline non-normality on the Shapiro-Wilk test but did not reach significance on the KS test or D'Agostino-Pearson test, suggesting approximate normality at the participant level. By this, response time analyses should employ log-transformed values, and non-parametric tests were considered as alternatives.

Method: Vigilance performance was analysed to verify attentiveness during encoding. A pie chart showing the overall pass/fail distribution, a grouped bar chart comparing pass rates between AB and NB

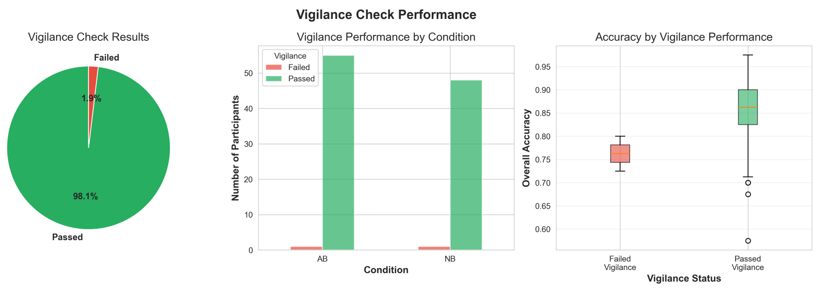


Figure 4: Vigilance Check Performance

conditions, and a violin plot comparing recognition accuracy between vigilance-pass and vigilance-fail participants.

Interpretation: The pie chart confirms that 159 of 161 participants (98.8%) passed the vigilance check. This substantially reduces the risk that inattentive encoding confounds. The grouped bar chart comparing vigilance pass rates between AB and NB conditions shows both conditions at near-ceiling, with pass rates visually indistinguishable. It confirms that the boundary manipulation did not differentially affect attentiveness during encoding.

6 Research Questions

6.1 H1: Overall Recognition Accuracy

Formal Hypothesis:

- **H_0:** There is no difference in overall recognition accuracy between NB and AB groups.
- **H_1:** NB participants show **higher** overall recognition accuracy than AB participants.

Rationale:
Event Segmentation Theory (EST) [1] proposes that continuous experience is parsed into discrete events at points of increased prediction error. These **event boundaries** serve as natural anchoring points in memory encoding. The Event Horizon Model [2] further posits a **boundary advantage**—event boundaries facilitate encoding into long-term memory because the cognitive processing at these transitions updates working memory's event models.

[4] demonstrated that disrupting natural event boundaries (e.g., inserting commercials at non-boundary points in a film) **impaired subsequent recall**, while boundaries preserved memory. [5] further showed that items present at event boundaries were recognised more readily, consistent with enhanced retrieval fluency at boundary-tagged memories.

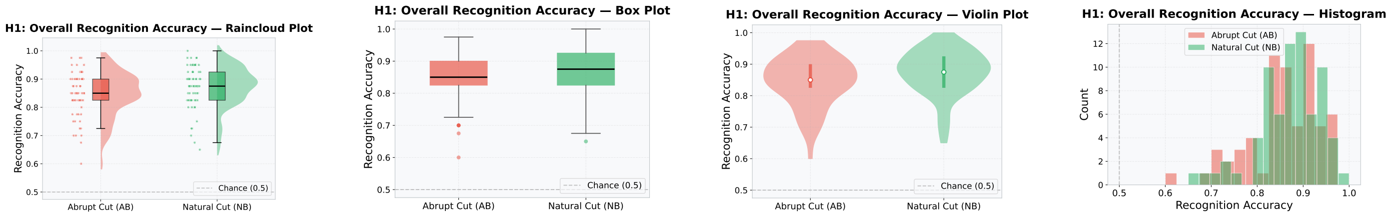
In the current experiment, the **Natural Boundary (NB)** condition preserves the original event timeline, while the **Abrupt Boundary (AB)** condition interrupts videos 1–5 seconds before a consensus boundary and resumes at a new event onset, thereby disrupting the natural flow of event segmentation.

Plot Details: The Y-axis represents overall recognition accuracy (proportion correct), calculated as $\text{acc}_i = \frac{1}{n_i}\sum_{j=1}^{n_i} \mathbf{1}[\text{resp.corr}_j = 1]$ across all $n_i = 40$ trials per participant. The X-axis represents the experimental condition (AB vs NB). The distributions are visualised using a raincloud plot, box plot, violin plot, and a histogram (15 bins) with a chance line indicated at 0.5.

Inference: The visualizations indicate a distinct advantage for the Natural Boundary (NB) condition in overall accuracy. Specifically, the violin and raincloud plots reveal that while NB scores are tightly clustered near the ceiling, the Abrupt Boundary (AB) distribution exhibits a pronounced lower tail. The box plot further confirms a lower median and greater interquartile range for the AB group.

6.2 H2: Before-Boundary (BB) Frame Accuracy

Formal Hypothesis:



(a) Raincloud (b) Box Plot (c) Violin Plot (d) Histogram

Figure 5: H1: Overall Recognition Accuracy — AB vs NB.

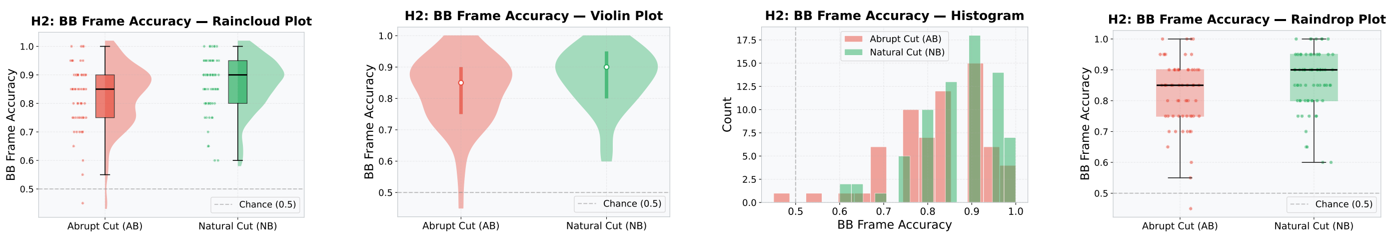
- **H$_0$:** No difference in BB-frame recognition accuracy between NB and AB groups.
- **H$_1$:** NB participants show **higher** BB-frame accuracy, reflecting preserved event continuity facilitating boundary-anchored encoding.

Rationale:
The **boundary advantage** is a central prediction of the Event Horizon Model [2]: event boundaries serve as anchoring points in long-term memory. Frames occurring just before a boundary should be well-encoded because the cognitive system is actively processing the transition between events.

[3] showed that deleting time segments *between* events preserved recall, while deleting segments *from* boundaries impaired it, demonstrating that boundary regions carry critical memory information.

In our experiment, **BB (Before-Boundary) frames** are target images extracted from time points just before an annotated event boundary. If abrupt cuts disrupt natural boundary processing, BB-frame recognition should be *specifically* impaired in the AB condition.

Plot Details: The Y-axis represents Before-Boundary (BB) frame accuracy, calculated as $\mathrm{acc}_i^{\mathrm{BB}} = \frac{1}{n_i^{\mathrm{BB}}} \sum_{j \in \mathrm{BB}} \mathbf{1}[\mathrm{resp.corr}_j = 1]$ for the approximately $n_i^{\mathrm{BB}} \approx 20$ BB trials per participant. The X-axis represents the condition (AB vs NB). The distributions are visualised using a raincloud plot, violin plot, a histogram (15 bins, chance line at 0.5), and a raindrop plot.



(a) Raincloud (b) Violin Plot (c) Histogram (d) Raindrop Plot

Figure 6: H2: Before-Boundary (BB) Frame Accuracy — AB vs NB.

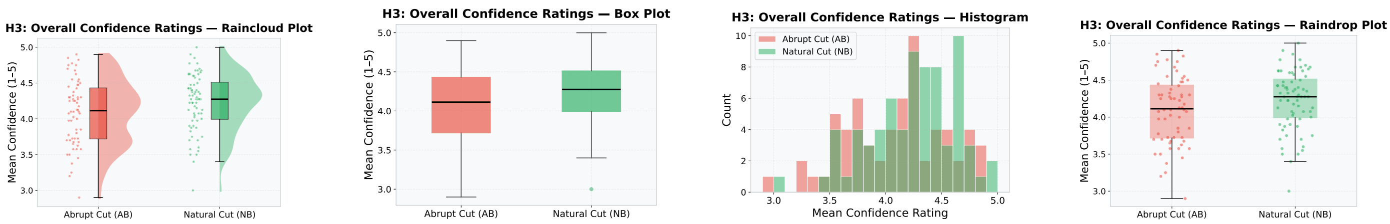
Inference: The data support a boundary-specific encoding advantage. The violin and raincloud plots demonstrate a higher density of top-tier accuracy scores for the NB group on BB frames. Conversely, the AB group's raindrop and histogram plots display increased spread and density in the lower accuracy ranges, indicating that premature video termination significantly degrades memory traces for temporally critical, boundary-adjacent information.

6.3 H3: Overall Confidence Ratings

Formal Hypothesis:

- **H$_0$:** No difference in mean confidence ratings between NB and AB groups.
- **H$_1$:** NB participants report **higher** confidence ratings, reflecting stronger memory traces from intact event continuity.

Rationale:
Natural event boundaries allow the cognitive system to properly consolidate event models [1], creating highly robust memory anchors [2]. Abruptly cutting a video disrupts this consolidation [4], producing fragmented and weaker memory traces. Because subjective confidence reflects the underlying strength of a memory trace, NB participants should experience greater retrieval fluency and report higher overall confidence than AB participants.
Plot Details: The Y-axis represents the mean confidence rating, calculated as the arithmetic mean of the 1–5 Likert scale responses ($\text{conf}_i = \frac{1}{n_i}\sum_{j=1}^{n_i} \text{conf\_radio.response}_j$). The X-axis represents the condition (AB vs NB). The distributions are visualised using a raincloud plot, box plot, a histogram (20 bins), and a raindrop plot.



(a) Raincloud (b) Box Plot (c) Histogram (d) Raindrop Plot

Figure 7: H3: Overall Confidence Ratings — AB vs NB.

Inference: Subjective confidence distributions align with objective accuracy. The box and raincloud plots show the NB group exhibits a positive shift in median confidence relative to the AB group. Additionally, the histogram reveals a higher frequency of responses clustered near the absolute ceiling (5.0) for NB participants. This indicates that intact event continuity facilitates stronger, more fluent memory representations, thereby enhancing metacognitive certainty during retrieval.

6.4 H4: Proportion of Low-Confidence Trials

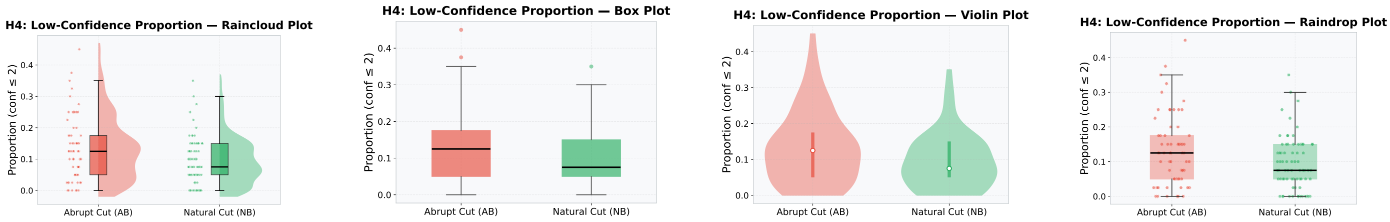
Formal Hypothesis:

- **H_0:** No difference in the proportion of low-confidence trials (confidence ≤ 2) between AB and NB groups.
- **H_1:** AB participants produce a **higher** proportion of low-confidence trials, reflecting degraded memory traces and increased uncertainty.

Rationale:
Interrupting a narrative at unnatural points prevents the cognitive system from effectively updating the event model [1], leading to impaired and degraded recall [4]. Without the natural transitions necessary for strong memory anchoring [2], participants are forced to rely on vague guesses rather than detail-rich retrieval, manifesting as a higher proportion of low-confidence trials in the AB condition.
Plot Details: The Y-axis represents the proportion of low-confidence trials, calculated as the fraction of trials per participant where confidence was rated as 1 or 2 ($\text{prop\_lc}_i = \frac{|\{j:\text{conf}_j \leq 2\}|}{n_i}$). The X-axis represents the condition (AB vs NB). The distributions are visualised using a raincloud plot, box plot, violin plot, and raindrop plot.
Inference: The AB condition is associated with a visibly higher proportion of low-confidence responses. As shown in the box and violin plots, the NB distribution is tightly anchored near zero, whereas the AB group exhibits a higher median and an extended upper tail reaching above 0.3. This variance, visible as an upward spread in the raindrop plot, suggests that disrupted event updating forces greater reliance on weak, familiarity-based recognition rather than confident recollection.

6.5 H5: Movie-Level Accuracy Variability

Formal Hypothesis:

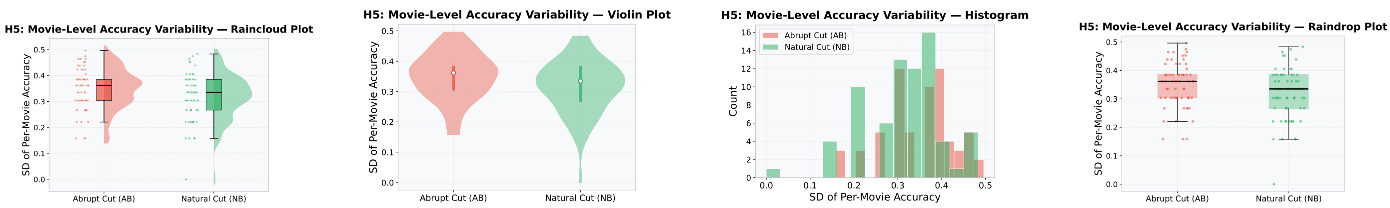


(a) Raincloud (b) Box Plot (c) Violin Plot (d) Raindrop Plot

Figure 8: H4: Proportion of Low-Confidence Trials — AB vs NB.

- $\mathbf{H_0}$: No difference in the variability (SD) of accuracy across movies between AB and NB groups.
- $\mathbf{H_1}$: AB participants show **higher** variability in per-movie accuracy, reflecting differential disruption severity across cuts.

Rationale:
Event boundaries provide the cognitive system with the necessary time to consolidate event models into long-term memory [1, 2]. In the Natural Boundary (NB) condition, intact transitions allow all participants to uniformly and properly encode the event, resulting in consistent performance and low accuracy variability (SD). In contrast, the Abrupt Boundary (AB) condition abruptly interrupts the video, depriving viewers of this crucial consolidation period [3]. Without a reliable structural anchor, different individuals encode the exact same movie in vastly different, fragmented ways, driving up the variability of accuracy scores across AB participants.
Plot Details: The Y-axis represents the standard deviation of per-movie accuracy, capturing the variability of performance across different videos for each participant ($\mathrm{SD}_i = \sqrt{\frac{1}{M-1}\sum_{m=1}^{M}(\mathrm{acc}_{i,m} - \bar{\mathrm{acc}}_i)^2}$). The X-axis represents the condition (AB vs NB). The distributions are visualised using a raincloud plot, violin plot, a histogram (15 bins), and a raindrop plot.



(a) Raincloud (b) Violin Plot (c) Histogram (d) Raindrop Plot

Figure 9: H5: Movie-Level Accuracy Variability — AB vs NB.

Inference: Per-movie accuracy variance is notably elevated in the AB condition. The raincloud and violin plots show the NB group's standard deviation is tightly clustered at a low baseline, reflecting uniform encoding success. In contrast, the AB group's histogram and raindrop plots reveal a rightward shift and wider dispersion. This implies that the severity of memory impairment is heterogeneous, varying significantly depending on the specific visual and narrative context of the interrupted video.

7 Future works

Future analytical phases will establish a rigorous inferential statistical pipeline to formally test the proposed hypotheses. To mitigate the characteristic positive skew of the response time data, a logarithmic transformation will be applied to all response times prior to analysis. Given the previously established violations of normality, primary two-group comparisons will utilize the non-parametric Mann-Whitney U test. For analyses requiring parametric frameworks (e.g., t-tests or ANOVAs), the assumption of homogeneity of variance will first be evaluated using Levene's test; any violations will be addressed by employing Welch's t-test or robust ANOVA alternatives. To assess main effects and interactions across multiple levels—specifically the mixed design of Condition (AB vs. NB) $\times$ Frame Type (BB vs. EM) a Friedman test coupled with Mann-Whitney follow-ups will be conducted. To strictly control for the multiple comparisons problem, appropriate alpha-level corrections (e.g., Bonferroni or Holm) will be applied. All post-hoc non-parametric results will be comprehensively reported, detailing the U-statistic, Z-score, p-value, and the rank-biserial correlation for effect size. Finally, target/lure discrimination will be formally modeled using Signal Detection Theory. The sensitivity index ($d' = Z(\text{Hit Rate}) - Z(\text{False Alarm Rate})$) and response criterion bias ($c = -0.5 \times [Z(\text{Hit Rate}) + Z(\text{False Alarm Rate})]$) will be computed for each participant and subsequently compared.

8 Team details

- Rajendraprasad S - Research Questions and Future Works
- Sathvika Miryala - Exploratory Data Analysis, Data cleaning and stats
- Sreemae Akhashathala - Exploratory data Analysis

9 Data Availability

All data, scripts, figures, and results are available in our replication package: `https://github.com/miryala10sathvika/brsm_movie_memory`

References

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[5] K. M. Swallow, J. M. Zacks, and R. A. Abrams, "Event boundaries in perception affect memory encoding and updating," *Journal of Experimental Psychology: General*, vol. 138, no. 2, pp. 236–257, 2009.